[1]
✓ 0s

```python
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
```

[8]
✓ 0s

```python
data = pd.read_csv("decisiontree2.csv")
x= data[["tempMode","AQ","USS","CS","VOC","RP","IP","Temperature"]]
y= data["fail"]
x
```

	tempMode	AQ	USS	CS	VOC	RP	IP	Temperature
0	7	7	1	6	6	36	3	1
1	1	3	3	5	1	20	4	1
2	7	2	2	6	1	24	6	1
3	4	3	4	5	1	28	6	1
4	7	5	6	4	0	68	6	1
...	...	...	...	...	...	...	...	...
939	7	7	1	6	4	73	6	24
940	7	5	2	6	6	50	6	24
941	3	6	2	7	5	43	6	24
942	6	6	2	5	6	46	7	24

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decisiontree2.ipynb

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Files

sample_data

decisiontree2.csv

[1]

```python
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
```

[7]

```python
data = pd.read_csv("decisiontree2.csv")
x= data[["tempMode","AQ","USS","CS","VOC","RP","IP","Temperature"]]
y= data["fail"]
data
```

	footfall	tempMode	AQ	USS	CS	VOC	RP	IP	Temperature	fail
0	0	7	7	1	6	6	36	3	1	1
1	190	1	3	3	5	1	20	4	1	0
2	31	7	2	2	6	1	24	6	1	0
3	83	4	3	4	5	1	28	6	1	0
4	640	7	5	6	4	0	68	6	1	0
...	...	...	...	...	...	...	...	...	...	...
939	0	7	7	1	6	4	73	6	24	1
940	0	7	5	2	6	6	50	6	24	1
941	0	3	6	2	7	5	43	6	24	1
942	0	6	6	2	5	6	46	7	24	1

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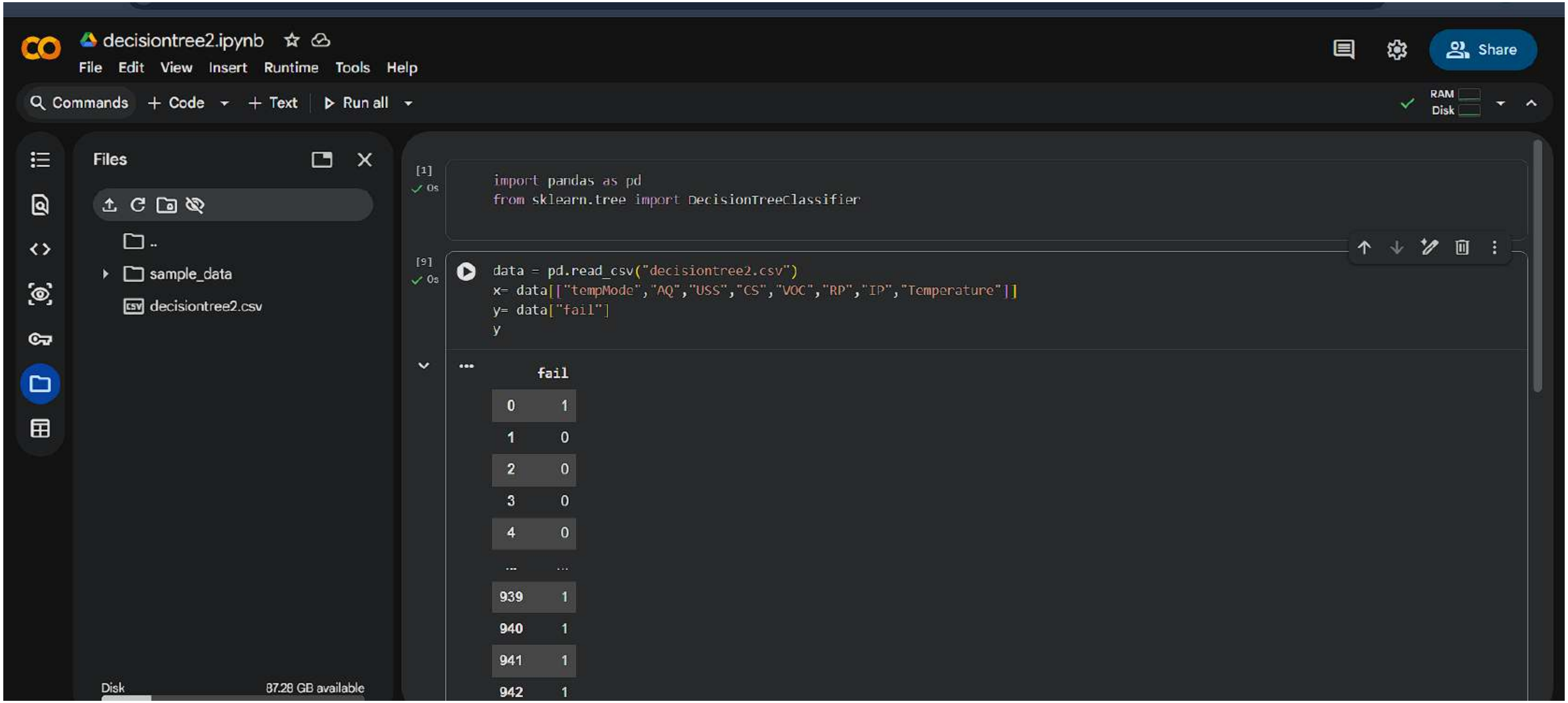
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sample_data
decisiontree2.csv
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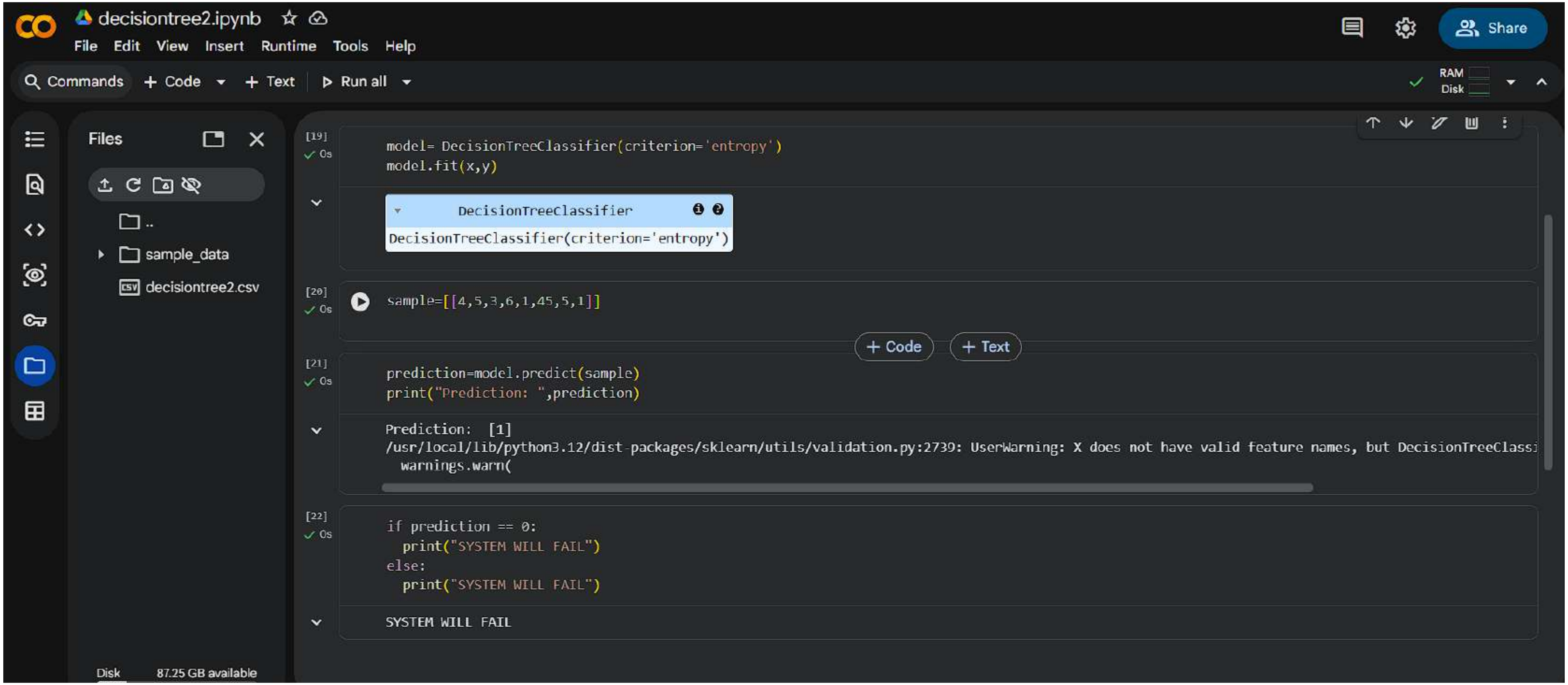
[1]
0s
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
[9]
0s
data = pd.read_csv("decisiontree2.csv")
x= data[["tempMode","AQ","USS","CS","VOC","RP","IP","Temperature"]]
y= data["fail"]
y
fail
0 1
1 0
2 0
3 0
4 0
... ...
939 1
940 1
941 1
942 1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decisiontree2.csv
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[19]
model= DecisionTreeClassifier(criterion='entropy')
model.fit(x,y)
DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy')
[20]
sample=[[4,5,3,6,1,45,5,1]]
+ Code + Text
[21]
prediction=model.predict(sample)
print("Prediction: ",prediction)
Prediction: [1]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassi
warnings.warn(
[22]
if prediction == 0:
print("SYSTEM WILL FAIL")
else:
print("SYSTEM WILL FAIL")
SYSTEM WILL FAIL